



Departamento de
Economía
Universidad de Concepción



Are fishing decisions flexible?

Participation, target, and landing choices in the U.S. West Coast pelagic fishery

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NOAA CAFA (NA20OAR4310507) ■ ANID-Chile FONDECYT Iniciación N° 11250223 ■ INCAR² CIA250009.

Motivation — spatial trade-offs run through the species portfolio

- Fishable area shrinking (**wind, conservation, mining**); species moving (**climate**).
- What matters is how fleets **redistribute effort** — follow, switch species, switch ports, halt, exit.
- Location choice usually models **one fishery at a time** $\Rightarrow$ the **species portfolio** drops out.

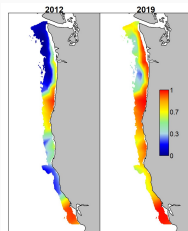
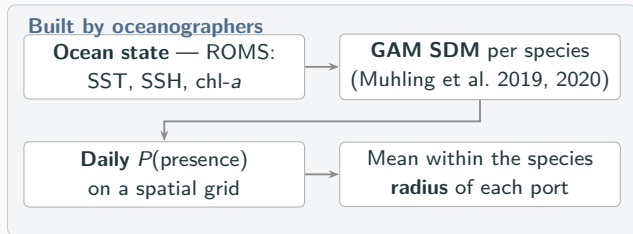
Question

How do **availability, prices, costs, weather** shape **daily** choices to **participate, target, land**?

2013–2017: the **2015 sardine closure** and the **2014–2016 heatwave**.

Where the boats end up is produced by **which species they can switch to** — so we model *participate* $\times$ *target* $\times$ *land* **jointly**.

The contribution — an oceanographic SDM as the economic regressor



Illustrative — Chasco et al. (2022)

$Avail_{p,s,t}$: an **exogenous**, gap-free measure of availability for
each port $\times$ species $\times$ day $\Rightarrow$ **enters the fisher's utility directly**

Assumption: habitat suitability tracks **potential catch rates** — if only partly, the availability effect is attenuated.

The usual proxy is **past catch**. Replacing it with an SDM is what makes the behavioral model **identifiable** and **climate-ready**.

Why that swap matters — past catch vs. SDM

Past catch (the usual proxy)

- ✗ **Endogenous** — where vessels already *chose* to fish.
- ✗ **Selection bias** — only at fished sites (Chen et al., 2023).
- ✗ **Gaps** — undefined when nobody fishes.

SDM habitat index (ours)

- ✓ **Exogenous** — an environmental input.
- ✓ **No gaps** — a value for *every* port–day.
- ✓ **Counterfactual-ready** — move the field, re-predict.

The third point buys us the **scenario** — you cannot shift a “past catch” field.

Setting — four fleet segments, four portfolios

U.S. West Coast Coastal Pelagic Species fishery: squid, sardine, anchovy, mackerels (+ non-CPS crab, salmon, tuna); purse seine. Segments from cluster analysis (Quezada et al., 2024).

Fleet segment	Main target(s)	Trips N	Vessels/yr
S. CCS industrial squid spec.	Market squid	29,160	42
Roving squid–sardine gen.	Squid & sardine	6,806	14
PNW sardine specialist	Pacific sardine	2,581	6
S. CCS forage-fish diverse	Diverse CPS	2,481	5

Same shock, **different portfolios** — the unit of adaptation is the *fleet segment*.

The model — nested logit over *participate* × *target* × *port*

Utility of alternative $j = (p, s, l)$ for vessel i on day t :

$$U_{itj} = \delta_j + \beta_{1,s} \text{Avail}_{tj} + \beta_{2,g} \text{Price}_{tj} + \dots + \varepsilon_{itj}$$

- **Avail** = the daily **SDM** index from the previous slide.
- Also: price, distances, wind, closures, unemployment.
- **State dependence** (last 30 days) — processor ties / port capacity.
- Nests: **participation**, then **species** — except forage-diverse, which nests by **port**.
- Crab/salmon split off (costly gear switch).
- Apollo (R); dominant species = target (**93.6%** of revenue).

Availability enters at **daily** frequency and is **exogenous** to the vessel's choice.

What drives the choice? Availability coefficients by segment

Availability of...	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Market squid	+2.45***	+1.75***	—	+1.61*
Northern anchovy	+1.35***	+4.19***	+1.16***	−1.88***
Pacific sardine	−1.78**	0.23	−0.59**	0.23
Chub & jack mackerel	−3.94***	−1.48	0.87	−1.92***
Non-CPS (crab/salmon)	—	+5.74***	+10.3***	—
Expected price (CPS/tuna)	0.26	+4.60***	+3.21***	+0.72***
State dependence (30-day)	+2.63***	+2.86***	+2.26***	+3.09***

Wind, distance-to-catch, distance-to-CG all (−); unemployment (+). Negative anchovy for forage-diverse: anchovy–squid availability correlate $r = -0.29$. *Read signs and significance*: coefficients are in utility units, so magnitudes are not comparable across segments — magnitude comes from the scenario (elasticities in backup).

Habit is **strong** (state dependence ≈ 2.3 – 3.1), yet the **SDM signal survives it** — so the flexibility we measure is genuine, not inertia.

Two adaptation modes — “on the move” vs. “in place”

Adapt *on the move* (species nests)

- Low $\lambda \Rightarrow$ strong correlation across **ports**, same species.
- Anchovy $\lambda = 0.49$; sardine $\lambda = 0.40$.
- $\Rightarrow$ **follow the species across ports.**

Adapt *in place* (port nests)

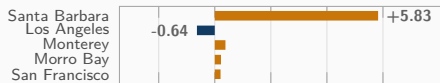
- Forage-diverse nests by **port**: Santa Barbara $\lambda = 0.49$ vs. LA/Monterey 0.73.
- Substitution across **species within a port.**
- $\Rightarrow$ **switch species, stay put.**

The mode is a **structural property of the fleet** (sensu Samhuri et al., 2024) — and it predicts who absorbs a **closed area**: “on the move” fleets re-route, “in place” fleets cannot.

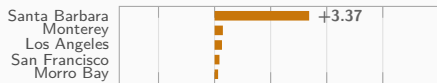
The payoff — move the habitat field, see where the effort goes

Squid habitat swapped for a *same-season* later window (centroid $42.99^\circ\text{N} \rightarrow 40.66^\circ\text{N}$); all else fixed.

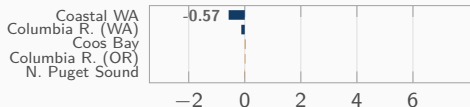
S. CCS squid specialist



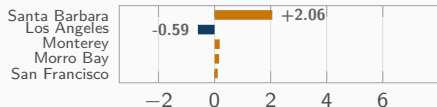
Roving squid-sardine



PNW sardine specialist



S. CCS forage diverse



$\Delta P(\text{landing at each port}), \text{pp}$: ■ gains effort ■ loses effort

Δ no-participation: $-6.0 / -4.2 / +0.6 / -1.9 \text{ pp}$ ■ same run as Figs. 5–6 of the paper, here by port.

Santa Barbara absorbs the squid. Los Angeles still nets -0.6 : it *gains* squid (+1.5) but **loses more chub mackerel** (-1.9) — the port total hides a **species recomposition**.

Takeaways

1. **Oceanographic SDMs work as economic regressors.** An exogenous, gap-free daily availability index replaces the endogenous past-catch proxy — and out-of-sample pseudo- ρ^2 is **0.22–0.44**.
2. **Harvesters are flexible.** Participation, target and landing all respond to availability, prices, costs and weather — *even* controlling for strong state dependence.
3. **The spatial answer runs through the portfolio.** Two adaptation modes — “on the move” (across *ports*) vs. “in place” (switch *species* within a port) — and Los Angeles shows why it matters: squid targeting there *rises* (+1.5 pp), yet the port **loses** effort (−0.6) because mackerel falls more.

Availability is an **ocean-model output**, so the same machinery runs forward under **climate projections**.



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Thank you! / ¡Gracias?

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Quezada-Escalona et al. (2026), *Ecological Economics* 247

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Backup ■ Descriptives — persistence *and* switching

Fleet segment	Species		Ports	
	Switches	Unique	Switches	Unique
S. CCS industrial squid specialist	19.8	2.6	20.7	3.3
Roving squid–sardine generalist	4.6	1.7	9.5	2.7
PNW sardine specialist	3.8	2.4	4.7	2.1
S. CCS forage-fish diverse	10.8	2.4	5.1	2.0

Vessel-level means, 2013–2017. Squid specialist: **switch rate 0.36, median run 2 days** — persistence *and* regular switching.

Short-run **persistence** + frequent **switching** $\Rightarrow$ recent choices matter (motivates state dependence).

Backup ■ Elasticities — price drives *entry*, availability *targeting*

Extensive margin (participate)

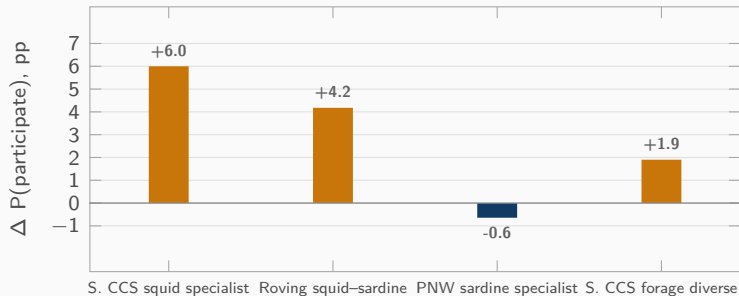
- Price dominates for mobile fleets: squid **1.71** (roving); anchovy **0.43** (PNW).
- Non-CPS availability matters where crab/salmon are relevant: crab **0.24**, sockeye **0.19** (PNW).

Intensive margin (conditional target)

- Availability often $>$ price: anchovy **2.55**, crab **2.12** (roving); crab **3.85**, sockeye **3.71** (PNW).
- Several $> 1 \Rightarrow$ more-than-proportional responses.
- Exception: forage-diverse is **price-driven**.

Prices move participation; **availability** moves which species, once at sea.

Backup ■ Scenario — change in *participation*

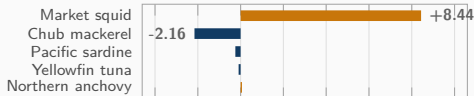


Change in the probability of **participating** (i.e. $-\Delta P(\text{no participation})$) under the southward squid redistribution. Generated from `scenario_dynamic_same_months_delta_participation_pp.csv`.

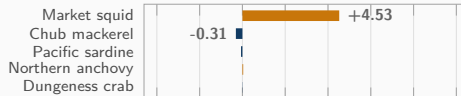
All three southern, squid-facing fleets go fishing **more often** when squid moves toward them; the PNW sardine fleet is essentially unaffected.

Backup ■ Scenario — change by *target species*

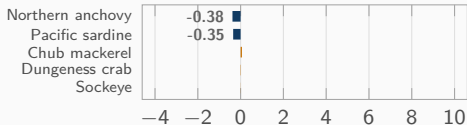
S. CCS squid specialist



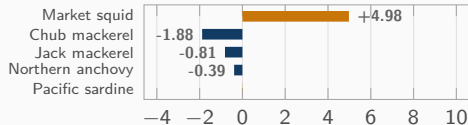
Roving squid–sardine



PNW sardine specialist



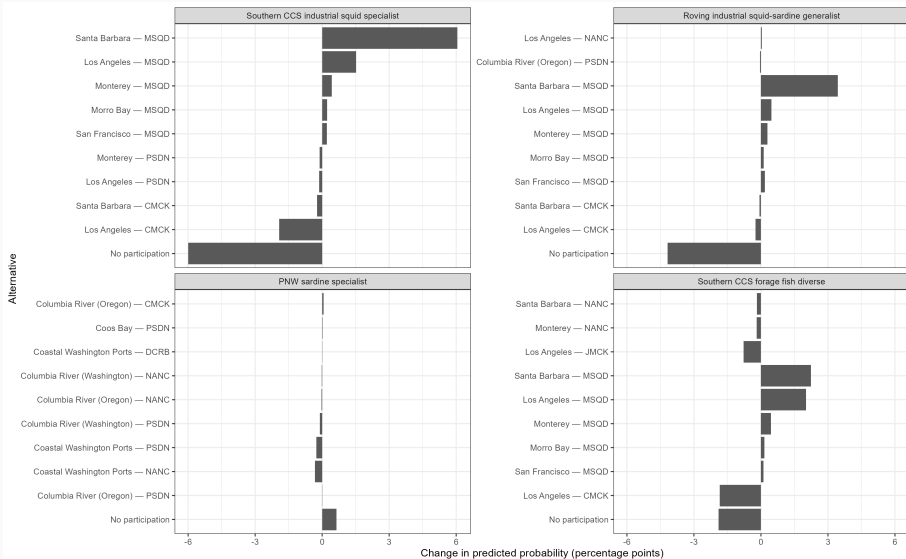
S. CCS forage diverse



△ predicted probability (pp): ■ more likely targeted ■ less likely ■ common axis.

Southern fleets **follow the squid** (+8.4 and +5.0 pp); the PNW sardine fleet barely moves — it has no squid to follow.

Backup ■ Scenario — change by port × species alternative



Backup ■ Dissimilarity parameters (λ)

Nest	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Participation (CPS/Tuna)	0.977	0.752	0.532	0.730
Market squid	0.777	0.752	—	—
Pacific sardine	0.617	0.403	0.404	—
Northern anchovy	0.494	—	0.532	—
Santa Barbara (port)	—	—	—	0.490
Los Angeles / Monterey (port)	—	—	—	0.730

Lower λ = stronger within-nest correlation (more substitution within the nest).

Backup ■ Variable construction

- **Availability:** binomial GAM SDMs (Muhling et al., 2019, 2020); mean presence within a species radius (60 km sardine/anchovy/mackerel, 90 km squid). CPUE z-score → sigmoid for non-SDM species.
- **Price:** predicted from $\ln(\text{price}) = \alpha + T_y + \gamma_m + \gamma_p$ (year trend, month & port FE); robustness with 30-day moving average.
- **Costs:** Haversine distance port→catch area and port→center of gravity.
- **Weather:** max ERA5 surface wind within 220 km. **Opportunity cost:** state unemployment.

Backup ■ Limitations

- **Aggregation bias:** port-area resolution may mask finer spatial heterogeneity (Dépalle et al., 2021).
- **Choice-set truncation:** rare alternatives folded into the outside option ($\geq 0.5\%$ filter); IIA assumed for omitted alternatives.
- **Price endogeneity:** predicted / MA prices reduce but do not eliminate simultaneity $\Rightarrow$ interpret price coefficients as **reduced-form** behavioral responses.
- **SDM proxy:** habitat suitability $\neq$ catch rate exactly $\Rightarrow$ availability effect may be attenuated.

Main qualitative result — **availability drives targeting** — is robust across price specifications.